LESSON 9.4
REFLECTIONS ON THE COORDINATE PLANE

LESSON INTRODUCTION

MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and a coordinate plane.

LEARNING TARGETS

- I can perform a reflection on the coordinate plane.
- I can describe relationships between figures after a reflection.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformed figure on a coordinate plane.

ESSENTIAL QUESTIONS

- How are the preimage and image related in a reflection?
- How do you perform a reflection on the coordinate plane?

LESSON NARRATIVE

Students develop the skills to perform a reflection on the coordinate plane. Students are asked to identify the coordinates of the image and the line of reflection.

COMPONENT SUMMARY

9.4.1 WARM-UP *(~5 MIN)*

Describe math used to remodel bathroom

9.4.2 GUIDED INSTRUCTION *(20 MIN)*

Discover how to perform reflection on coordinate plane

9.4.3 COLLABORATION *(5 MIN)*

Discuss which quadrant image of reflection will be in

9.4.4 YOUR TURN! *(10 MIN)*

Practice performing reflections on coordinate plane

9.4.5 WRAP-UP *(~5 MIN)*

Reflect on mastery of learning targets

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Reflection
- Line of Reflection

MATERIALS

- (Optional) Graph Paper
- (Optional) Patty Paper

ADDITIONAL PREPARATION

- Enlarged visual representations of figures could be created on card stock and cut out prior to lesson.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to find the coordinates, the image, and the line of reflection when the preimage and image overlap.
- Students may forget to mirror the shape when performing a reflection.

THINGS TO CONSIDER

- Consider avoiding having students memorize a formula or relationship. Students in grade 8 should not be using algebraic notations or ordered pair rules. That is reserved for Geometry. Instead, focus on the relationship that students define using their own words as well as that all points in the image are equidistant from the line of reflection as the point in the preimage.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

In 9.4.2 Guided Instruction, ask students what they notice about the coordinates of the corresponding vertices of the preimage and image. Then, prompt them to share what they wonder about the relationship between the coordinates. This discussion will help students build confidence in and understanding of how to perform reflections on the coordinate plane.

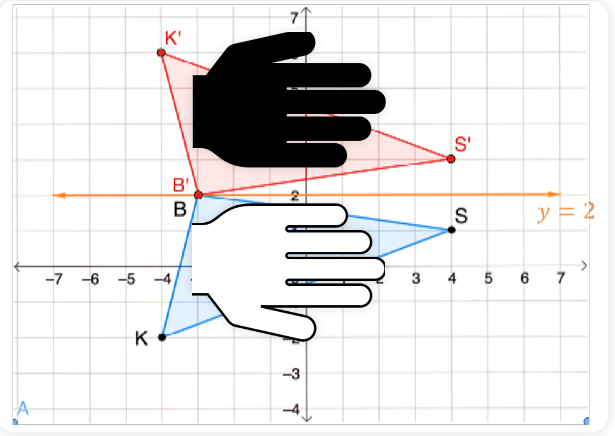
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

The Warm-Up prompts students to describe the mathematical concepts involved in remodeling a bathroom. Consider asking students if they have done any renovations at their house or built anything. Student interest will be ignited when they see the utility of the math they are learning.

DIFFERENTIATE INSTRUCTION

Consider providing a kinesthetic entry-point by having students place their hand on the graph lined up with the line of reflection and flip their hand across the line of reflection. This will help student see why the preimage is flipped across the line of reflection to create the image. Describing the reflections verbally will reinforce student understanding and provide an auditory entry-point.



9.4.1 WARM-UP: THINK LIKE A MATHEMATICIAN

TEACHER MOVES

This Warm-Up prompts students to describe the mathematical concepts involved in remodeling a bathroom. Consider asking students if they have done any renovations at their house or built anything.

9.4.1 Warm-Up: Think Like a Mathematician

1. The image shows the beginning of a bathroom remodel.

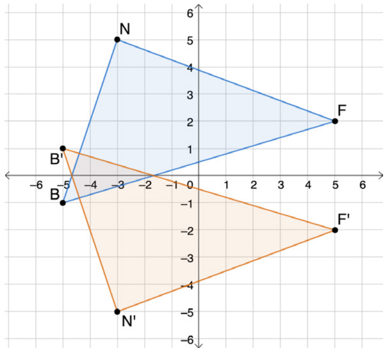
List some of the mathematics that may be needed to complete a remodel such as this one. Include at least three things on your list.

Answers will vary. Some mathematics involved in a remodel might include measuring distances, finding the area of a space to be covered, and finding the perimeter of a space.

9.4.2 GUIDED INSTRUCTION: REFLECTIONS ON THE COORDINATE PLANE

9.4.2 Guided Instruction: Reflections on the Coordinate Plane

1. On the coordinate plane shown, ΔNBF is reflected to create $\Delta N'B'F'$.



   a. What is the line of reflection?
   The x-axis or the line $y = 0$.

   b. Complete the table with the coordinates of each vertex.

Vertex	Coordinates	Vertex	Coordinates
Point N	$(-3, 5)$	Point N'	$(-3, -5)$
Point B	$(-5, -1)$	Point B'	$(-5, 1)$
Point F	$(5, 2)$	Point F'	$(5, -2)$

TEACHER MOVES

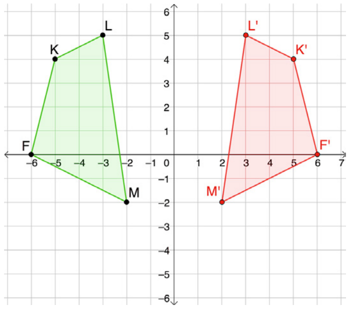
Students may struggle to find the coordinates of the image and the line of reflection when the preimage and image overlap. Remind students that each corresponding point should be the same distance from the line of reflection. Prompt students to reflect the points one at a time.

QUESTIONING

Challenge students who finish early to also reflect the triangle NBF over the y-axis.

   c. What do you notice about the coordinates of the corresponding vertices in the preimage and the image?
   Answers may vary. I notice that the x-coordinate of the corresponding vertices is the same, and the y-coordinate is the opposite.

2. Quadrilateral $FKLM$ is shown on the coordinate grid.



   a. Complete the left side of the table for points F, K, L, and M.

Vertex	Distance From y-axis	Coordinates	Vertex	Distance From y-axis	Coordinates
F	6	$(-6, 0)$	F'	6	$(6, 0)$
K	5	$(-5, 4)$	K'	5	$(5, 4)$
L	3	$(-3, 5)$	L'	3	$(3, 5)$
M	2	$(-2, -2)$	M'	2	$(2, -2)$

   b. Reflect quadrilateral $FKLM$ across the y-axis. Label the image quadrilateral $F'K'L'M'$.

   c. Complete the right side of the table in Part A for vertices F', K', L', and M'.

ENRICHMENT

Notice and Wonder

Ask students what they notice about the coordinates of the corresponding vertices of the preimage and image. Then, prompt them to share what they wonder about the relationship between the coordinates. This discussion will help students build confidence in and understanding of how to perform reflections on the coordinate plane.

DIFFERENTIATE INSTRUCTION

Seeing the reflections on the coordinate plane provides a visual entry-point. Describing the reflections verbally will reinforce student understanding and provide an auditory entry-point.

9.4.2 GUIDED INSTRUCTION: REFLECTIONS ON THE COORDINATE PLANE (CONTINUED)

ENRICHMENT

What would your answer to question 2d be if the line of reflection was the x-axis?

DIFFERENTIATE INSTRUCTION

Consider providing a kinesthetic entry-point by having students place their hand on the graph lined up with the line of reflection and flip their hand across the line of reflection. This will help student see why the preimage is flipped across the line of reflection to create the image.

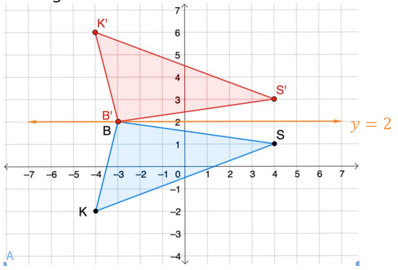
QUESTIONING

Challenge students to generalize how a reflection over the x-axis impacts coordinates (x,y). Then, prompt them to generalize how a reflection over the y-axis impacts coordinates (x,y).

d. Describe what you notice about the coordinates of the corresponding vertices in the preimage and the image.
Answers will vary. I notice that the y-coordinate of the corresponding vertices is the same, and the x-coordinate is the opposite.

3. Figures can be reflected across any line on the coordinate plane, not only the x- or y-axis.

Triangle SBK and the line $y = 2$ are shown on the coordinate grid.



a. Reflect ΔSBK across the line $y = 2$. Label the image $\Delta S'B'K'$.

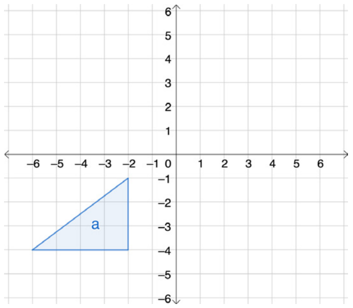
b. Complete the table with the coordinates of the vertices of the preimage and image.

Vertex	Coordinates	Vertex	Coordinates
Point S	$(4, 1)$	Point S'	$(4, 3)$
Point B	$(-3, 2)$	Point B'	$(-3, 2)$
Point K	$(-4, -2)$	Point K'	$(-4, 6)$

c. Describe the relationship between the coordinates of points B and B'.
Answers will vary. The coordinates of B and B' are the same because they are on the line of reflection.

9.4.3 Collaboration: Which Quadrant?

1. A triangle is shown on the coordinate plane.



a. Without performing the transformation, discuss with your partner what the image of the triangle would look like after a reflection across the y-axis.

b. Complete the statement.

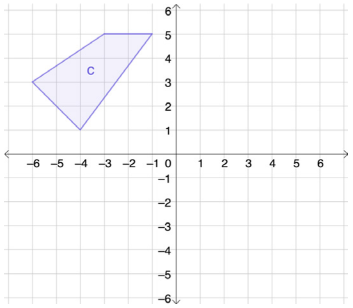
The preimage of the triangle is in quadrant

- 1.
- 2.
- 3.
- 4.

The image of the triangle after a reflection across the y-axis would lie in quadrant

- 1.
- 2.
- 3.
- 4.

2. A quadrilateral is shown on the coordinate plane.



Without performing the transformation, complete the statement.

The preimage of the quadrilateral is in quadrant

- 1.
- 2.
- 3.
- 4.

The image of the quadrilateral after a reflection across the x-axis would lie in quadrant

- 1.
- 2.
- 3.
- 4.

9.4.3 COLLABORATION: WHICH QUADRANT?

DIFFERENTIATE INSTRUCTION

Prompt students to label each quadrant before answering the questions to provide a visual cue and help prevent mistakes.

TEACHER MOVES

Poll the Class

Consider having students hold up their fingers to indicate which quadrant the image will be in. Discuss incorrect responses to resolve misunderstandings.

DIFFERENTIATE INSTRUCTION

Prompt students to label the axes to help prevent students from accidentally reflecting the figure over the incorrect axis.

9.4.4 YOUR TURN!

ENRICHMENT

- What do you notice about the coordinates of the preimage and image?
- What would the coordinates of the image be if the reflection was over the y-axis?

DIFFERENTIATE INSTRUCTION

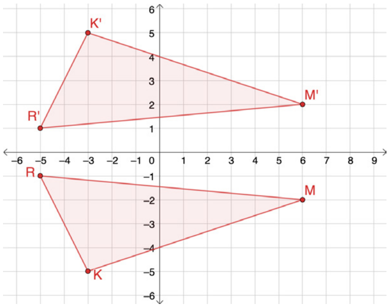
If students struggle to find the line of reflection when shapes overlap, consider having them imagine the shapes as wings on a butterfly and ask where they would draw the body (center of the butterfly).

9.4.4 Your Turn!

1. The vertices of ΔKMR have their coordinates given in the table.

Vertex	Coordinates	Vertex	Coordinates
Point K	$(-3,-5)$	Point K'	$(-3,5)$
Point M	$(6,-2)$	Point M'	$(6,2)$
Point R	$(-5,-1)$	Point R'	$(-5,1)$

a. Plot ΔKMR on the coordinate grid.

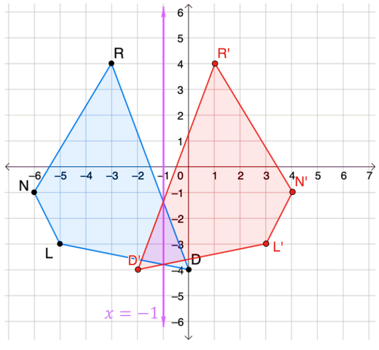


b. Reflect ΔKMR across the x-axis. Label the image $\Delta K'M'R'$.

c. Complete the table above for the image.

d. Describe the relationship between the corresponding vertices of the preimage and image.
Answers will vary. The x-coordinates of the corresponding vertices are the same, and the y-coordinates are opposites.

2. Quadrilateral $RDLN$ and line $x = -1$ are shown on the coordinate grid.



a. Reflect quadrilateral $RDLN$ over the line of reflection, $x = -1$. Label the image $R'D'L'N'$.

b. Describe how each pair of corresponding vertices relate to each other compared to the line of reflection.

Corresponding Vertices	Description of Relationship
R and R'	*They are both 2 units from the line of reflection.*
D and D'	*They are both 1 unit from the line of reflection.*
L and L'	*They are both 4 units from the line of reflection.*
N and N'	*They are both 5 units from the line of reflection.*

9.4.5 Wrap-Up: Reflection

Complete each statement as you reflect on today's learning.

1. When someone asks me what I did in math today, I can say …

   Answers will vary.

2. The thing that made the most sense to me today was …

   Answers will vary.

3. Something that I do not understand ***YET*** is …

   Answers will vary.

9.4.5 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

The data provided from this formative assessment, coupled with what was observed during the lesson, can help you differentiate instruction in subsequent lessons.